Summary of Completed Analyses & Missing Questions

Based on the three PDFs I was able to read, here's what has been thoroughly investigated and what important questions remain:

What's Been Analyzed

1. Peer Co-Occurrence Dynamics (2 pages)

Key Finding: Strong evidence of **episode-wide regime changes** with synchronized winning patterns.

• Simultaneous wins are common (peak at 3 machines winning together)
• **Critical discovery**: Average payout increases monotonically with synchronization level (1.18 for solo wins → 2.37 for 8-machine synchronization)
• Directly validates the "shared_change" condition hypothesis

2. Machine Sub-Archetypes (3 pages)

Key Finding: Machines exhibit **bimodal distributions** within nominal types, with opposing momentum polarities.

• **Balanced machines**: Clear split between mean-reverting (negative momentum, 45-46% win rate) and momentum-driven (positive momentum, declining win rates)
• **Steady machines**: Heavily momentum-driven with "hot hand" behavior (win rates 0.54 → 0.84 with momentum)
• **Jackpot machines**: Subtle bimodal pattern with lower overall win rates
• **Important**: No temporal reversal detected—machine archetypes remain fixed throughout episodes
• Momentum index from early history (first 20-50 ticks) remains predictive

3. Short_History Bottleneck Analysis (4 pages)

Key Finding: Model performance gap (R=0.6242) can be improved with three specific features.

Priority discoveries:

• **Peer payout coupling**: U-shaped relationship for balanced/jackpot (peak at medium peer payouts), linear for steady machines
• **Jackpot cold streak hazard**: High payout rate collapses 4× after 9-10 consecutive zeros (19% → 4.7%)
• **Minimal recency drift**: Recent 40 ticks vs older 24 ticks show stable distributions—error is from sample size, not drift
• **No hard budget ceiling** on floor-wide payouts, but peer levels predict target behavior non-linearly

Critical Missing Analyses

Based on your mention of three other PDFs and gaps in the current analyses, here are the **most important unanswered questions**:

1. Empirical Transition Probabilities (mentioned PDF missing)

What we need:

• Full transition matrices for each machine type and condition
• State-to-state transition probabilities (0x → 1x, 1x → 2x, etc.)

• How do transitions differ between conditions (stable vs drift vs shared_change)?
• Are there absorbing states or cyclic patterns?

Why it matters: Understanding transition dynamics is crucial for sequential prediction and could explain the condition-specific performance differences.

2. Drift Hazard & Drought Prediction (mentioned PDF missing)

What we need:

• How do "drought" periods (extended zero-payout streaks) evolve in the drift condition?
• What triggers the end of droughts?
• Are there early warning signals for regime shifts?
• How does drift differ from stable in terms of hazard rates over time?

Why it matters: The "drift" condition showed gradual improvement in late stages (200+ ticks) but we don't understand the mechanism.

3. 3-State Empirical Transitions (mentioned PDF missing)

What we need:

• What are the three states? (Likely: losing/neutral/winning or low/medium/high payout)
• Transition probabilities between these states
• How do machine types behave differently in each state?
• State duration distributions and escape probabilities

Why it matters: Simplifying to 3 states could reveal clearer patterns than analyzing all 5 payout levels separately.

4. Slot Machine Prediction Model (PDF exists but couldn't be read)

What we need:

• Model architecture and features used
• Performance metrics by condition
• Feature importance rankings
• Where the model fails (error analysis)

Why it matters: Understanding what's already been tried prevents duplication and reveals where to focus next.

=Additional High-Value Questions Not Yet Addressed

5. Cross-Episode Patterns

• Do certain episodes systematically outperform others?
• Can we predict episode difficulty from early-tick behavior?
• Are there "episode types" beyond the four conditions?

6. Machine Learning Opportunity Analysis

• Which features from the completed analyses have been incorporated into models?
• What's the incremental value of each discovered pattern?

• Can we build ensemble models that specialize by condition?

7. Temporal Cascade Effects

• When one machine hits a jackpot, how do peer machines respond in the next 5-10 ticks?
• Are there "contagion" effects where wins spread across the floor?
• Do machines "compete" for payouts or "cooperate"?

8. Optimal Strategy Development

• Given limited history, which machines should be played?
• When should a player switch machines vs. persist?
• How does the optimal strategy differ by condition?

9. Condition Classification

• Can we identify which condition we're in from early observations?
• What's the minimum number of ticks needed to classify the condition?
• Are there hybrid conditions or transitions between conditions?

=R̂ecommended Next Steps

Immediate priorities (if you can share the missing PDFs):

1. Read the three missing PDFs to complete the picture
2. Synthesize transition probabilities with momentum patterns
3. Build a unified feature set combining all discoveries

If those PDFs aren't available, I recommend:

1. **Validate the peer coupling hypothesis** with fresh queries on the data
2. **Build transition matrices** for each condition and machine type
3. **Analyze the shared_change 200+ tick spike** in detail—this is a dramatic anomaly
4. **Investigate episode-level predictors** to understand why some episodes perform 50% better than others

Would you like me to dive deep into any of these specific areas using the available data?